

First consider the special case where $X = e$; let $\bar{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$ and let $\bar{e} = \text{Spec}(\bar{\mathbb{F}}_p)$. The sheaf F is entirely determined by its fibre $F_{\bar{e}}$, regarded as a finite-dimensional Ω -vector space on which the Galois group

$$\pi_e = \text{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_p)$$

acts continuously. Since π_e is topologically generated by the Frobenius element

$$f : \lambda \mapsto \lambda^p \quad (\lambda \in \bar{\mathbb{F}}_p),$$

the data of the sheaf F is equivalent to that of the vector space $F_{\bar{e}}$ endowed with an automorphism $f_{F_{\bar{e}}}$ which is “topologically unipotent”, that is, whose k -th power tends to the identity in the analytic group $\text{Aut}(F_{\bar{e}})$ as k tends “multiplicatively” to infinity. We set

$$L_F(t) = 1/\det(1 - f_{F_{\bar{e}}}^{-1}t) \in \Omega[[t]].$$

Now suppose that $X = \text{Spec}(\mathbb{F}_q)$, with $q = p^d$ (d an integer ≥ 1). Let $\bar{x}$ be a geometric point of X , defined by an extension of $\mathbb{F}_q$ into $\bar{\mathbb{F}}_p$; the Galois group

$$\pi_x = \text{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_q)$$

is identified with the subgroup of π_e topologically generated by f^d . Thus the data of the sheaf F is equivalent to that of its fibre $F_{\bar{x}}$, a finite-dimensional vector space over Ω , endowed with a topologically unipotent automorphism $(f^d)_{F_{\bar{x}}}$, which we write as $f_{F_{\bar{x}}}^d$ for simplicity. Let g be the unique morphism from X to e ; the direct image $g_*(F)$ is likewise defined by its fibre $(g_*F)_{\bar{e}}$ with the operations of π_e , which is computed as an “induced module”

$$(g_*F)_{\bar{e}} \simeq F_{\bar{x}} \otimes_{\pi_x} \pi_e.$$

We set

$$L_F(t) = L_{g_*F}(t) = 1/\det(1 - f_{(g_*F)_{\bar{e}}}^{-1}t);$$

an easy determinant calculation shows that this expression is equal to

$$L_F(t) = 1/\det(1 - f_{F_{\bar{x}}}^{-d}t^d).$$

In the general case of a scheme X of finite type over e , endowed with a constructible Ω -sheaf F , define

$$L_F = \prod_{x \in X^0} L_{F_x},$$

where X^0 denotes the set of closed points of X ; we shall see that this product is multipliable in $\Omega[[t]]$. Indeed, let us make it explicit by introducing, for each $x \in X^0$, a geometric point $\bar{x}$ of X localized at x , defined by an extension of the residue field $k(x)$ into $\bar{\mathbb{F}}_p$, and by setting

$$d(x) = [k(x) : \mathbb{F}_p]$$

(the degree of x). With this notation

$$L_{F_x}(t) = 1/\det(1 - f_{F_{\bar{x}}}^{-d(x)}t^{d(x)}) = 1 + \text{Tr}(f_{F_{\bar{x}}}^{-d(x)}t^{d(x)}) + \dots$$

Since for every integer n the set of points $x \in X^0$ such that $d(x) \leq n$ is finite, it follows that the family $(L_{F_x})_{x \in X^0}$ is multipliable in $\Omega[[t]]$. Thus the L -function of (X, F) is

$$L_F(t) = \prod_{x \in X^0} 1/\det(1 - f_{F_{\bar{x}}}^{-d(x)}t^{d(x)}).$$

The first formal properties of the L -function are collected in the following statement.

Proposition 1.

a) Multiplicativity in F . Let

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

be an exact sequence of constructible Ω -sheaves on X ; then

$$L_F = L_{F'} L_{F''}.$$

b) Let Y be a closed subscheme of X and let $U = X - Y$ be its open complement. Then

$$L_F = L_{F|U} L_{F|Y}$$

for every constructible Ω -sheaf F .

c) Let $X \rightarrow S$ be a morphism of schemes of finite type over $\mathbb{F}_p$, and let F be a constructible Ω -sheaf on X ; then

$$L_F = \prod_{s \in S^0} L_{F|X_s}.$$

Assertion a) is clear; b) follows from $X^0 = U^0 \cup Y^0$ as a disjoint union, and c) follows from

$$X^0 = \prod_{s \in S^0} X_s^0$$

using associativity of multipliable infinite products. These formulas on closed points are valid because the image of a closed point under a morphism of preschemes of finite type over $\mathbb{F}_p$ is again a closed point.

2. Statement of the rationality theorem and reduction to the case of a curve

Theorem. Let X be a scheme of finite type over $\mathbb{F}_p$, of dimension n , let $g : X \rightarrow e$ be its structural morphism, and let F be a constructible Ω -sheaf on X . Then

$$L_F = \prod_{i=0}^{2n} (L_{R_!^i g(F)})^{(-1)^i}. \quad (1)$$

In this statement the $R_!^i g(F)$ are the higher direct images with proper support, in the sense of ℓ -adic cohomology. We put

$$H_!^i(\bar{X}, \bar{F}) = R_!^i \bar{g}(\bar{F}) = R_!^i g(F)_{\bar{e}}$$

(by the base-change theorem), where $\bar{X}$, $\bar{F}$, and $\bar{g}$ are obtained from X , F , and g by the base change $\bar{e} \rightarrow e$, with $\bar{e}$ the spectrum of an algebraic closure of $\mathbb{F}_p$. With this notation, by no. 1,

$$L_{R_!^i g(F)}(t) = 1 / \det(1 - f_{H_!^i(\bar{X}, \bar{F})}^{-1} t) = L_{f_{H_!^i(\bar{X}, \bar{F})}^{-1}}(t),$$

where $L_u(t) = 1/\det(1-ut)$ for every endomorphism u of a finite-dimensional vector space, according to the formalism developed in an earlier exposé.

More generally, consider a morphism $u : X \rightarrow Y$ of schemes of finite type over $\mathbb{F}_p$ and a constructible Ω -sheaf F on X . Let $(P_{F,u})$ denote the following property:

$$(P_{F,u}) : \quad L_F = \prod_i (L_{R_{\dagger}^i u(F)})^{(-1)^i}.$$

The theorem for the pair (X, F) is precisely $(P_{F,g})$, where $g : X \rightarrow e$ is the structural morphism of X . We shall reduce it to the case where X is the projective line by means of a few lemmas.

Lemma 1. Let $u : X \rightarrow Y$ be a morphism of schemes of finite type over $\mathbb{F}_p$, and let F be a constructible Ω -sheaf. If $(P_{F|X_y, u_y})$ is true for every $y \in Y^0$, then $(P_{F,u})$ is true.

Indeed,

$$L_F = \prod_{y \in Y^0} L_{F|X_y}$$

(no. 1, Proposition 1c), and hence

$$= \prod_{y \in Y^0} \prod_i (L_{R_{\dagger}^i u_y(F|X_y)})^{(-1)^i}$$

using $(P_{F|X_y, u_y})$; but

$$R_{\dagger}^i u_y(F|X_y) \simeq R_{\dagger}^i u(F)_y$$

by the base-change theorem, which gives

$$L_F = \prod_i \left(\prod_{y \in Y^0} L_{(R_{\dagger}^i u(F))_y} \right)^{(-1)^i} = \prod_i (L_{R_{\dagger}^i u(F)})^{(-1)^i}$$

by the definition of the L -function on Y .

Corollary. Let $u : X \rightarrow Y$ be a finite morphism of schemes of finite type over $\mathbb{F}_p$, and let F be a constructible Ω -sheaf on X . The property $(P_{F,u})$ is true.

The lemma allows us to reduce to the case where Y is punctual: $Y = \text{Spec}(\mathbb{F}_q)$. Then X is finite and

$$u_*(F) \simeq \bigoplus_{x \in X} u_*(F_x).$$

To establish the formula $(P_{F,u})$, namely $L_F = L_{u_* F}$, one is therefore reduced to the case where X itself is punctual: $X = \text{Spec}(\mathbb{F}_q)$. Let $g : X \rightarrow e$ and $h : Y \rightarrow e$ be the structural morphisms. By definition (no. 1),

$$L_F = L_{g_* F} = L_{h_* u_* F} = L_{u_* F},$$

since $g = h \circ u$.

Lemma 2. Let $u : X \rightarrow Y$ and $v : Y \rightarrow Z$ be morphisms of schemes of finite type over $\mathbb{F}_p$, and let F be a constructible Ω -sheaf on X . If the properties $(P_{F,u})$ and $(P_{R_{\dagger}^i u(F), v})$ for all i are true, then $(P_{F, v \circ u})$ is true.

Indeed,

$$\begin{aligned} L_F &= \prod_i (L_{R_!^i u(F)})^{(-1)^i} = \prod_i \prod_j (L_{R_!^j v(R_!^i u(F))})^{(-1)^i (-1)^j} \\ &= \prod_{i,j} (L_{R_!^j v(R_!^i u(F))})^{(-1)^{i+j}} \end{aligned}$$

by $(P_{F,u})$ and $(P_{R_!^i u(F),v})$. The Leray spectral sequence

$$R_!^j v(R_!^i u(F)) \implies R_!^k (v \circ u)(F)$$

and the multiplicativity of L_F with respect to F (no. 1, Proposition 1a) finally give

$$L_F = \prod_k (L_{R_!^k (v \circ u)(F)})^{(-1)^k},$$

as required.

Lemma 3. Let X be a scheme of finite type over $\mathbb{F}_p$, and let $g : X \rightarrow e$ be its structural morphism.

- a) Let F be a constructible Ω -sheaf on X , let Y be a closed subscheme of X , and let $U = X - Y$ be the open complement. If the theorem is true for two of the three pairs (X, F) , $(Y, F|_Y)$, $(U, F|_U)$, then it is true for the third.
- b) Let

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

be an exact sequence of constructible Ω -sheaves on X . If the theorem is true for two of the three pairs (X, F') , (X, F) , (X, F'') , then it is true for the third.

We shall use only a), but we also give b) for symmetry. The proofs of a) and b) are analogous; let us prove a), for example. The exact sequence

$$\cdots \longrightarrow R_!^i g_U(F|_U) \longrightarrow R_!^i g(F) \longrightarrow R_!^i g_Y(F|_Y) \longrightarrow R_!^{i+1} g_U(F|_U) \longrightarrow \cdots$$

and Proposition 1a) (no. 1) give

$$\prod_i (L_{R_!^i g(F)})^{(-1)^i} = \prod_i (L_{R_!^i g_U(F|_U)} L_{R_!^i g_Y(F|_Y)})^{(-1)^i},$$

and Proposition 1b) (no. 1) then allows one to conclude.

The arguments of Lemmas 1, 2, and 3 may be summarized by saying that, by Proposition 1 of no. 1, the function L_F has formal properties analogous to those of cohomology with proper support.

We now use the preceding lemmas to reduce to the case where X is the projective line.

First reduction: by noetherian induction one may construct a partition $(U_j)_{1 \leq j \leq r}$ of X by affine schemes such that U_{j+1} is open in $X - \bigcup_{i=1}^j U_i$ for $j = 1, \dots, r-1$. Suppose that the theorem has been established for every pair (U, G) , where U is an affine scheme of finite type over $\mathbb{F}_p$ and G is a constructible Ω -sheaf on U . It is then true for $(U_j, F|_{U_j})$, $j = 1,$

ldots, r ; by Lemma 3a), descending induction on j shows that it is true for $(\bigcup_{i \geq j} U_i, F|_{\bigcup_{i \geq j} U_i})$,

and hence for (X, F) , since $X = \bigcup_{i \geq 1} U_i$. It is therefore enough to prove the theorem in the case where X is affine.

Second reduction: suppose X affine of dimension n . There exists a finite morphism $u : X \rightarrow \mathbb{E}^n$ from X into numerical space of dimension n . We reason by induction on n . For $n = 0$, X is finite over e and the theorem is true by the corollary to Lemma 1. If $n \geq 1$, suppose the theorem established for affine schemes of dimension $n - 1$; let $u : X \rightarrow \mathbb{E}^n$ be a finite morphism and let $v : \mathbb{E}^n \rightarrow \mathbb{E}^1$ be a projection morphism, defined from an isomorphism $\mathbb{E}^n \simeq \mathbb{E}^1 \times \mathbb{E}^{n-1}$. Composing, one obtains a morphism $w = v \circ u : X \rightarrow \mathbb{E}^1$ whose fibres are affine schemes of dimension $n - 1$. Lemma 1 then shows that the property $(P_{F,w})$ is true by the induction hypothesis. By Lemma 2, it is enough to establish the theorem for the pairs $(\mathbb{E}^1, R_i^! w(F))$. Thus one is reduced to the case where $X = \mathbb{E}^1$ is the affine line.

Third reduction: suppose $X = \mathbb{E}^1$. Let $i : \mathbb{E}^1 \rightarrow \mathbb{P}^1$ be the immersion of the affine line into the projective line $\mathbb{P}^1$, and let $Z = \mathbb{P}^1 - i(\mathbb{E}^1)$ be the reduced closed subscheme at the point at infinity of $\mathbb{P}^1$. Since the theorem is true for $(Z, i_!(F)|_Z)$, it is equivalent, by Lemma 3a), to establish it for $(\mathbb{E}^1, F)$ or for $(\mathbb{P}^1, i_!(F))$; since i is an open immersion one has $F = i_!(F)|_{\mathbb{E}^1}$.

3. Proof of the theorem

After these reductions, it remains to establish the theorem in the case where X is the projective line. We shall first transform formula (1) of the statement into a Lefschetz-type formula involving traces; this latter formula will be proved in the particular case where X is a projective smooth curve.

Return to formula (1):

$$L_F = \prod_i (L_{R_i^! g(F)})^{(-1)^i} = \frac{P_F^1 \cdots P_F^{2n-1}}{P_F^0 \cdots P_F^{2n}}, \quad \text{where} \quad P_F^i(t) = \det(1 - f_{H_i^1(\bar{X}, \bar{F})}^{-1} t).$$

The logarithm of the left-hand side is

$$\sum_{x \in X^0} \sum_{m=1}^{\infty} \text{Tr}(f_{F_{\bar{x}}}^{-md(x)}) t^{md(x)} / m = \sum_{n=1}^{\infty} \left(\sum_{d(x)|n} d(x) \text{Tr}(f_{F_{\bar{x}}}^{-n}) \right) t^n / n,$$

while that of the right-hand side is

$$\sum_{n \geq 1} \left(\sum_i (-1)^i \text{Tr}(f_{H_i^1(\bar{X}, \bar{F})}^{-n}) \right) t^n / n.$$

Comparing the coefficients of t^n/n , one sees that it is necessary to establish

$$\sum_{d(x)|n} d(x) \text{Tr}(f_{F_{\bar{x}}}^{-n}) = \sum_i (-1)^i \text{Tr}(f_{H_i^1(\bar{X}, \bar{F})}^{-n}) \quad (2)$$

for all $n \geq 1$; these calculations are legitimate because the field Ω has characteristic zero.

Transform formula (2) further by supposing that X is proper, which allows us to write ordinary cohomology instead of cohomology with proper support. Recall that the category of $\mathbb{Q}_\ell$ -sheaves on X is a category of fractions of the category of ℓ -adic sheaves. By abuse of notation, we shall henceforth denote by $F = (F_\nu)_{\nu \in \mathbb{N}}$ the ℓ -adic sheaf on X corresponding to the $\mathbb{Q}_\ell$ -sheaf underlying F . Then $\bar{F} = (\bar{F}_\nu)$ and the Frobenius morphisms $\text{Fr}_{\bar{F}_\nu/X}^*$ define a morphism of ℓ -adic sheaves

$$\text{Fr}_{\bar{F}/X}^* : \text{fr}_X^*(F) \longrightarrow \bar{F},$$

hence an ℓ -adic correspondence on $(\bar{X}, \bar{F})$. For each integer ν we know that the endomorphism $\text{fr}_{R\Gamma_{\bar{X}}(\bar{F}_\nu)}$ defined by $(\bar{\text{fr}}_X, \text{Fr}_{\bar{F}_\nu/X}^*)$ is the inverse of $f_{R\Gamma_{\bar{X}}(\bar{F})}$ (§2, no. 3, corollary to Proposition 3). Using the definition of ℓ -adic cohomology, one then finds that the endomorphism $\text{fr}_{H^i(\bar{X}, \bar{F})}$ of $H^i(\bar{X}, \bar{F})$ defined by $(\bar{\text{fr}}_X, \text{Fr}_{\bar{F}/X}^*)$ is the inverse of $f_{H^i(\bar{X}, \bar{F})}$; thus

$$f_{H^i(\bar{X}, \bar{F})}^{-n} = \text{fr}_{H^i(\bar{X}, \bar{F})}^n$$

for all $n \geq 1$ and all i .

Apply this result in the particular case where $X = x = \text{Spec}(\mathbb{F}_q)$ ($q = p^d$) is reduced to a point whose degree d divides n . We obtain

$$f_{F_{\bar{x}}}^{-n} = \text{fr}_{F_{\bar{x}}}^n = (\text{Fr}_{F_{\bar{x}}/X}^*)^n$$

($\bar{x}$ is the geometric point defined by an extension of $\mathbb{F}_q$ into $\bar{\mathbb{F}}_p$).

On the other hand, $\bar{\text{fr}}_X$ and f_X act in inverse ways on the points of $\bar{X}$. It follows that the projection $\bar{X} \rightarrow X$ establishes a bijective correspondence between the set of closed points of degree d of X and the set of d -element orbits of $\bar{\text{fr}}_X$ acting on $\bar{X}$. Noting that the set $\bar{X}^{\bar{\text{fr}}_X^n}$ of points of $\bar{X}$ fixed by $\bar{\text{fr}}_X^n$ is the union of the orbits of $\bar{\text{fr}}_X$ whose cardinality divides n , one can rewrite (2) in the “geometric” form

$$\sum_{\bar{x} \in \bar{X}^{\bar{\text{fr}}_X^n} \text{ rep.}} \text{Tr}(f_{F_{\bar{x}}}^n) = \sum_i (-1)^i \text{Tr}(\text{fr}_{H^i(\bar{X}, \bar{F})}^n). \quad (2')$$

This is a Lefschetz-type formula on the scheme $\bar{X}$ endowed with the sheaf $\bar{F}$ and the correspondence $(\bar{\text{fr}}_X, \text{Fr}_{\bar{F}/X}^*)$.

To lighten notation, we shall henceforth write X , F , h , and h_F instead of $\bar{X}$, $\bar{F}$, $\bar{\text{fr}}_X$, and $(\text{Fr}_{\bar{F}/X}^*)^n$; since X and F will no longer appear in their former sense, no confusion can arise. We shall suppose that X is a smooth projective curve over $k = \bar{\mathbb{F}}_p$. The fixed points of h on X all have multiplicity 1: indeed, h acts by $T \mapsto T^{p^n}$ on the completed local ring of a fixed point, where T is a uniformizer. The theorem will follow from the following result.

Proposition 2. Let X be a smooth projective curve over an algebraically closed field k of characteristic p , let F be a constructible ℓ -adic sheaf on X_{et} (with ℓ a prime number distinct from p), let h be an endomorphism of X whose fixed points all have multiplicity 1, and let $h_F : h^*(F) \rightarrow F$ be a morphism of sheaves. The following formula, where $h_{H^i(X, F)}$ denotes the endomorphism of $H^i(X, F)$ defined by (h, h_F) , is true:

$$\sum_{x \in X^h} \text{Tr}((h_F)_x) = \sum_i (-1)^i \text{Tr}(h_{H^i(X, F)}). \quad (2'')$$

In this seminar, an analogous Lefschetz formula has been established for a constructible torsion sheaf F_ν , annihilated by $\ell^{\nu+1}$ on the curve X , endowed with a morphism $h_{F_\nu} : h^*(F_\nu) \rightarrow F_\nu$. It is written

$$\sum_{x \in X^h} \text{Tr}((h_{F_\nu})_x) = \text{Tr}_{R\Gamma_X(F_\nu)}^*, \quad (3)$$

where both sides belong to $\mathbb{Z}/\ell^{\nu+1}\mathbb{Z}$.

Suppose that the ℓ -adic sheaf in the statement is written as $F = (F_\nu)_{\nu \in \mathbb{N}}$, with F_ν annihilated by $\ell^{\nu+1}$. For each ν we may write formula (3) $_\nu$, and one sees that the left-hand sides of these formulas,

as ν varies, are the components of an ℓ -adic integer which is precisely the left-hand side of formula (2'') to be proved. It remains to establish that the right-hand sides of the formulas $(3)_\nu$ define, as ν varies, an ℓ -adic integer equal to

$$\sum_i (-1)^i \operatorname{Tr}(h_{H^i(X, F)}), \quad \text{where} \quad H^i(X, F) = \varprojlim_\nu H^i(R\Gamma_X(F_\nu)).$$

This follows from purely algebraic lemmas that we shall now prove to finish.

Consider a projective system $(K_\nu)_{\nu \in \mathbb{N}}$ of perfect complexes, with $K_\nu \in D_{\text{parf}}(\mathbb{Z}/\ell^{\nu+1}\mathbb{Z})$, such that for each ν the transition morphism $K_{\nu+1} \rightarrow K_\nu$ is isomorphic, in the sense of derived categories, to the canonical morphism

$$K_{\nu+1} \rightarrow K_{\nu+1}/\ell^{\nu+1}K_{\nu+1}.$$

Suppose each complex K_ν is endowed with an endomorphism $h_\nu \in \operatorname{Fl}(D_{\text{parf}}(\mathbb{Z}/\ell^{\nu+1}\mathbb{Z}))$, so that $(h_\nu)_{\nu \in \mathbb{N}}$ is compatible with the transition morphisms, thereby defining an endomorphism of the projective system (K_ν) . In the application to the proof of Proposition 2 one takes

$$K_\nu = R\Gamma_X(F_\nu), \quad h_\nu = h_{R\Gamma_X(F_\nu)};$$

the condition on transition morphisms follows by the ‘‘K nneth formula’’ from the analogous condition verified by the transition morphisms of the ℓ -adic projective system (F_ν) .

We propose to find a projective system $(K'_\nu)_{\nu \in \mathbb{N}}$, where K'_ν is a complex of free $(\mathbb{Z}/\ell^{\nu+1}\mathbb{Z})$ -modules of finite type, zero outside an interval of degrees $[a, b]$ independent of ν , and endowed with an endomorphism of complexes $h'_\nu : K'_\nu \rightarrow K'_\nu$, such that the following conditions are satisfied:

- 1) the transition morphism $K'_{\nu+1} \rightarrow K'_\nu$ is isomorphic, as a morphism of complexes, to

$$K'_{\nu+1} \rightarrow K'_{\nu+1}/\ell^{\nu+1}K'_{\nu+1}$$

for every $\nu \in \mathbb{N}$;

- 2) $(h'_\nu)_{\nu \in \mathbb{N}}$ is compatible with the transition morphisms;
- 3) (K'_ν, h'_ν) is isomorphic to (K_ν, h_ν) in $D_{\text{parf}}(\mathbb{Z}/\ell^{\nu+1}\mathbb{Z})$ for every ν , these isomorphisms being choosable so as to be compatible with the transition morphisms of (K'_ν) and (K_ν) .

Assuming the construction of the systems (K'_ν) and (h'_ν) , we may write

$$\operatorname{Tr}^* h_\nu = \operatorname{Tr}^* h'_\nu = \sum_i (-1)^i \operatorname{Tr} h_\nu^i$$

for every $\nu \in \mathbb{N}$. Let

$$K' = \varprojlim_\nu K'_\nu, \quad h' = \varprojlim_\nu h'_\nu$$

be the induced endomorphism of K' . Then $(\operatorname{Tr}^* h_\nu)_\nu$ is an ℓ -adic integer equal to $\sum_i (-1)^i \operatorname{Tr} h'^i$. Moreover, the projective system (K'_ν) satisfies the Mittag-Leffler condition, and so does the system $(H^{i-1}(K'_\nu))_\nu$ for every i ; it follows that

$$H^i(K') \simeq \varprojlim_\nu H^i(K'_\nu)$$

for every i (EGA 0_{III} 13.2.3). In the case of interest to us, where $K_\nu = R\Gamma_X(F_\nu)$ and $h_\nu = h_{R\Gamma_X(F_\nu)}$, $H^i(K')$ is identified with $H^i(X, F)$, and the endomorphism of $H^i(K')$ defined by h' is identified with $h_{H^i(X, F)}$. Consequently

$$\sum_i (-1)^i \operatorname{Tr} h'^i = \sum_i (-1)^i \operatorname{Tr} h_{H^i(X, F)},$$

and formula (2'') in the statement is thereby proved.

The existence of the systems (K'_ν) and (h'_ν) rests on two lemmas, in which we consider a commutative ring A , an ideal I of A , and the functor

$$T : L \longmapsto L \otimes_A A_0$$

where L is a complex of A -modules bounded on the right, with $A_0 = A/I$.

Lemma 1. Suppose that the ideal I is nilpotent. Consider a complex $L \in D^-(A)$ flat in every degree, a complex $M \in D^-(A_0)$ projective in every degree, and an isomorphism

$$u : T(L) \xrightarrow{\sim} M$$

in $D^-(A_0)$. There exist a complex L' of A -modules, bounded on the right and projective in every degree, an isomorphism

$$v : L \xrightarrow{\sim} L'$$

in $D^-(A)$, and an isomorphism of complexes

$$w : T(L') \xrightarrow{\sim} M,$$

such that $u = w \circ T(v)$ in $D^-(A_0)$.

Remark. If M is assumed to be free, respectively of finite type, in every degree, then the complex L' is also free, respectively of finite type, in every degree. Indeed, if a flat A -module N is such that $T(N)$ is free, respectively of finite type, it is easy to show that N is free, respectively of finite type (Bourbaki, Commutative Algebra, Chap. II, §3, Prop. 5). Note the corollary: let $L \in \operatorname{ob} D^-(A)$ be such that $L \otimes_A^{\mathbb{L}} A_0$ is a pseudo-coherent, respectively perfect, complex of A_0 -modules, that is, isomorphic in $D^-(A)$ to a complex bounded above, respectively bounded, and projective of finite type in every degree. Then L is a pseudo-coherent, respectively perfect, complex of A -modules.

Lemma 2. Consider a complex L of A -modules bounded on the right, projective in every degree, endowed with an endomorphism φ , and an endomorphism φ_0 of $M = T(L)$ coinciding with $T(\varphi)$ in the sense of the derived category $D^-(A_0)$. There exists an endomorphism φ' of L which coincides with φ in the sense of $D^-(A)$ and satisfies $T(\varphi') = \varphi_0$.

Before proving these lemmas, let us indicate how they are applied to prove the existence of the systems (K'_ν) and (h'_ν) . We argue by induction on ν , supposing that $K'_0, h'_0, \dots, K'_{\nu-1}, h'_{\nu-1}$ have already been constructed. Apply the lemmas with

$$A = \mathbb{Z}/\ell^{\nu+1}\mathbb{Z}, \quad I = \ell^\nu \mathbb{Z}/\ell^{\nu+1}\mathbb{Z}$$

so that $I^2 = 0$, and

$$A_0 = \mathbb{Z}/\ell^\nu \mathbb{Z}.$$

First take $L = K_\nu$ and $M = K'_{\nu-1}$ in the first lemma. In view of the remark, this gives the existence of $K'_\nu = L'$ with transition morphism $K'_\nu \rightarrow K'_{\nu-1}$ given by w , so that condition 1) above is satisfied. The isomorphism v in $D^-(A)$ allows h_ν to be transported to an endomorphism of K'_ν in the sense of $D^-(A)$. We may suppose that the latter comes from an endomorphism of complexes $h''_\nu : K'_\nu \rightarrow K'_\nu$. Applying the second lemma with $L = K'_\nu$, $\varphi = h''_\nu$, and $\varphi_0 = h'_{\nu-1}$, we obtain the existence of $h'_\nu : K'_\nu \rightarrow K'_\nu$ satisfying condition 2); condition 3) is also satisfied. Note that, with the notation of Lemma 1, if $M^i = 0$ for an index i , then $L^i = 0$ follows from Nakayama's lemma; the complexes K'_ν are therefore indeed all zero outside one and the same interval of degrees.

To prove Lemmas 1 and 2 we shall need the following result.

Lemma 3.

- (i) Let P and Q be A -modules. If P is projective, then for every A_0 -linear map $f_0 : T(P) \rightarrow T(Q)$ there exists an A -linear map $f : P \rightarrow Q$ such that $T(f) = f_0$.
- (ii) Suppose that the ideal I is nilpotent. For every projective A_0 -module P_0 there exists a projective A -module P such that $T(P) \simeq P_0$. If moreover P' is a flat A -module such that $T(P') \simeq P_0$, then $P' \simeq P$.

Assertion (i) is clear: if u_P (resp. u_Q) denotes the canonical map from P onto $T(P)$ (resp. from Q onto $T(Q)$), then the composite $f_0 \circ u_P$ is an A -linear map from P to $T(Q)$, and u_Q is surjective. Since P is projective, there is a factorization of $f_0 \circ u_P$ through u_Q , namely $f_0 \circ u_P = u_Q \circ f$, where f is an A -linear map from P to Q ; then $T(f) = f_0$.

To prove (ii), first observe that the case where P_0 is free is evident, with no hypothesis on the ideal I . The general case, where P_0 is projective, is reduced to this by introducing a free A_0 -module L_0 of which P_0 is a direct factor, and a free A -module L such that $T(L) \simeq L_0$. The module P_0 is identified with the image of a projector $e_0 : L_0 \rightarrow L_0$, and it remains to see that there exists a projector $e : L \rightarrow L$ lifting e_0 . This follows from Bourbaki, Commutative Algebra, Chap. III, §4, no. 6, Lemma 2, applied to the sub- A -algebra B of $\text{End}(L)$ generated by an endomorphism of L lifting e_0 (whose existence is ensured by (i)) and to the nilpotent ideal which is the trace on B of the kernel of the canonical homomorphism $\text{End}(L) \rightarrow \text{End}(L_0)$. This proves the existence of P in (ii). If moreover P' is as in (ii), then the composite isomorphism $T(P) \simeq P_0 \simeq T(P')$ comes, by (i), from a homomorphism $f : P \rightarrow P'$. Since P and P' are flat over A and $T(f)$ is an isomorphism, it follows that $\text{Gr}_I(f)$ is an isomorphism; hence f is an isomorphism, since I is nilpotent. This proves (ii).

Corollary. Suppose the ideal I is nilpotent. For every acyclic complex N_0 of projective A_0 -modules, bounded on the right, there exists an acyclic complex N of projective A -modules, bounded on the right, such that $T(N) \simeq N_0$.

The complex N_0 , being acyclic, decomposes into short exact sequences

$$0 \longrightarrow Z_0^i \longrightarrow N_0^i \longrightarrow Z_0^{i+1} \longrightarrow 0,$$

where Z_0^i denotes, for each i , the submodule of cycles of N_0^i ; moreover $N_0^i = 0$ for all sufficiently large i . By descending induction on i , one sees that the preceding exact sequences are split and that Z_0^i is projective for all i . One lifts N_0 by lifting all these exact sequences step by step, by descending induction on i . It is therefore enough to show that if

$$S_0 : 0 \longrightarrow P_0 \longrightarrow Q_0 \longrightarrow R_0 \longrightarrow 0$$

is an exact sequence of A_0 -modules with Q_0 projective, and if R is an A -module such that $T(R) \simeq R_0$, then there exists an exact sequence

$$S : 0 \longrightarrow P \longrightarrow Q \longrightarrow R \longrightarrow 0$$

of A -modules, with Q projective, such that $T(S) \simeq S_0$. Now Q_0 can be lifted to a projective A -module Q by Lemma 3(ii), and $Q_0 \rightarrow R_0$ can be lifted to an A -linear map $Q \rightarrow R$ by Lemma 3(i). This map is surjective by Nakayama's lemma; one sets $P = \ker(Q \rightarrow R)$ to complete the exact sequence S .

Proof of Lemma 2: since $M = T(L)$ is projective in every degree, the hypothesis of the lemma says that $T(\varphi)$ is homotopic to φ_0 . Thus there exists a homotopy operator $k_0 = (k_0^i)$, with $k_0^i : M^i \rightarrow M^{i-1}$, such that

$$T(\varphi) - \varphi_0 = d_0 k_0 + k_0 d_0,$$

where d_0 is the differential of M . By Lemma 3(i), k_0^i lifts to a linear map $k^i : L^i \rightarrow L^{i-1}$. This gives a homotopy operator $k = (k^i)$ in L . It is enough to set

$$\varphi' = \varphi - kd - dk$$

where d is the differential of L , to verify Lemma 2.

Proof of Lemma 1: after replacing L by a quasi-isomorphic complex if necessary, we may suppose that the complex L , and hence also $T(L)$, is projective in every degree; we shall assume this henceforth. Thus we may suppose that the isomorphism u of $D^-(A_0)$ comes from a quasi-isomorphism $T(L) \rightarrow M$, again denoted by u . We shall first reduce to the case where u is surjective. The complex N_0 defined by

$$N_0^i = M^i \oplus M^{i-1}, \quad d^i(x, y) = (d^i x, x - d^{i-1} y)$$

($x \in M^i, y \in M^{i-1}$) is visibly acyclic and projective in every degree; an epimorphism of complexes $N_0 \rightarrow M$ is defined by the projection $N_0^i \rightarrow M^i$ in degree i . The corollary to Lemma 3 allows us to lift N_0 to an acyclic complex N , bounded on the right and projective in every degree. We may then replace L by $L \oplus N$, which is quasi-isomorphic to it, since N is acyclic, and projective in every degree. At the same time, replace u by the quasi-isomorphism

$$T(L \oplus N) = T(L) \oplus N_0 \longrightarrow M$$

defined by u on $T(L)$ and by the epimorphism $N_0 \rightarrow M$ on N_0 ; it is clear that this last quasi-isomorphism is surjective.

Suppose therefore that $u : T(L) \rightarrow M$ is a surjective quasi-isomorphism, and let R_0 be its kernel. The complex R_0 is acyclic because u is a quasi-isomorphism, and it is projective in every degree because the exact sequence

$$0 \longrightarrow R_0 \longrightarrow T(L) \longrightarrow M \longrightarrow 0$$

is split, since M is projective in every degree. Hence, by the corollary to Lemma 3, R_0 can be lifted to a complex R of A -modules, bounded on the right, acyclic, and projective in every degree. We now show that the morphism $f_0 : R_0 \rightarrow T(L)$ lifts to a morphism of complexes $f : R \rightarrow L$. For this, observe that R_0 is homotopically trivial, so that f_0 is homotopic to 0 and can be written

$$f_0 = h_0 d_R + d_{T(L)} h_0,$$

where $h_0 = (h_0^i)_i$, with $h_0^i : R_0^i \rightarrow T(L)^{i-1}$. Lemma 3(i) allows h_0 to be lifted to $h = (h^i)_i$, with $h^i : R^i \rightarrow L^{i-1}$, and one sees that

$$f = h d_R + d_L h$$

is a morphism of complexes lifting f_0 . A standard lemma (compare SGA 60/61, IV 5.7) then implies that f is injective and that $\text{coker } f = L'$ is flat in every degree. Since

$$T(L') \simeq \text{coker } T(f) = \text{coker } f_0 \simeq M$$

is projective in every degree, it follows, by Lemma 3(ii), that the same is true of L' . On the other hand, the canonical homomorphism $L \rightarrow L'$ is a quasi-isomorphism, being surjective with acyclic kernel R ; thus L' satisfies the desired conclusion of Lemma 1, q.e.d.

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